

CHARCOAT

PASSIVE FIRE PROTECTION

CharCoat IC



Insulator Primer

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THE USE OF AN INSULATOR COATING PRIMER

One of the important differences between CharCoat IC and other insulator coatings is that a separate CharCoat IC Primer material is supplied for spraying on to the insulator prior to the application of the coating. This provides important advantages as described below.

Silicone rubber does not naturally adhere to other materials – this characteristic being the reason for its widespread use in the manufacture of moulds, as the items are easily released once cured. This, however, is not a favourable property for coatings and a bonding agent has to be used to ensure that the coating is well attached to the insulator and no peeling is experienced. Most coating manufacturers prefer to simplify the coating application by mixing the bonding agent into the coating itself. This has two main disadvantages, these being:

- Only a small percentage of the bonding agent ends up on the insulator surface – the remainder remains within the coating layer and serves no useful purpose.
- Further, the presence of the bonding agent reduces the silicone content of the coating thus adversely affecting its performance.

With the use of CharCoat IC Primer, not only is the performance of the coating enhanced but the adhesion to the insulator is superior.

This is well illustrated by subjecting coating test specimens to the 12 cycle, 100 hour boiling test specified in Clause 6.6.1 of the IEEE Std. 1523 for insulator coatings.

As shown in the adjacent image, the specimen with the separate pre- applied primer achieves far better adhesion than when 10% by weight of bonding agent is added to the coating itself or when no primer or bonding agent is used.

The full test is described in bonding test data

Having the bonding agent separate and used as a primer, ensures that all of it is where it is needed, i.e., at the interface between the coating layer and the porcelain surface and, moreover, the coating, being free of any extraneous material serving no beneficial purpose, has a higher silicone content thus yielding excellent performance and a long life in extremely polluted environments.

